

SQL QUERIES

◆ 1. Order and Sales Analysis :-

A. Total number of orders by order_status

```
SELECT order_status, COUNT(*) AS total_orders  
FROM customers_orders  
GROUP BY order_status;
```

B. Total sales revenue by order_status

```
SELECT order_status, ROUND(SUM(order_amount), 2) AS total_sales  
FROM customers_orders  
GROUP BY order_status;
```

C. Monthly revenue trend

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,  
       ROUND(SUM(order_amount), 2) AS monthly_revenue  
FROM customers_orders  
GROUP BY month  
ORDER BY month;
```

D. Average order value by status

```
SELECT order_status, ROUND(AVG(order_amount), 2) AS avg_order_value
```

```
FROM customers_orders  
GROUP BY order_status;
```

◆ 2. Customer Analysis

A. Repeat vs. one-time customers

```
SELECT  
    CASE  
        WHEN order_count = 1 THEN 'One-time customer'  
        ELSE 'Repeat customer'  
    END AS customer_type,  
    COUNT(*) AS num_customers  
FROM (  
    SELECT customer_id, COUNT(*) AS order_count  
    FROM customers_orders  
    GROUP BY customer_id  
) AS sub  
GROUP BY customer_type;
```

B. Top 5 customers by total spend

```
SELECT customer_id, ROUND(SUM(order_amount), 2) AS total_spent  
FROM customers_orders  
GROUP BY customer_id  
ORDER BY total_spent DESC  
LIMIT 5;
```

C. Orders per customer over time

```
SELECT customer_id, DATE_FORMAT(order_date, '%Y-%m') AS month, COUNT(*) AS  
orders  
  
FROM customers_orders  
  
GROUP BY customer_id, month  
  
ORDER BY customer_id, month;
```

◆ 3. Payment Status Analysis

A. Payment status count

```
SELECT payment_status, COUNT(*) AS count  
  
FROM payments  
  
GROUP BY payment_status;
```

B. Failed payments by method

```
SELECT payment_method, COUNT(*) AS failed_count  
  
FROM payments  
  
WHERE payment_status = 'failed'  
  
GROUP BY payment_method;
```

C. Success rate by payment method

```
SELECT payment_method,  
       SUM(payment_status = 'completed') AS success_count,  
       SUM(payment_status = 'failed') AS fail_count,
```

```
ROUND(SUM(payment_status = 'completed') / COUNT(*), 2) AS success_rate
FROM payments
GROUP BY payment_method;
```

◆ 4. Order Details Report

A. Join customers_orders and payments for complete order info

```
SELECT
    o.order_id,
    o.customer_id,
    o.order_date,
    o.order_amount,
    o.order_status,
    o.shipping_address,
    p.payment_method,
    p.payment_status,
    p.payment_date,
    p.payment_amount
FROM customers_orders o
LEFT JOIN payments p ON o.order_id = p.order_id;
```

B. Orders with failed or pending payments

```
SELECT o.order_id, o.customer_id, o.order_date, p.payment_status
FROM customers_orders o
JOIN payments p ON o.order_id = p.order_id
WHERE p.payment_status IN ('failed', 'pending');
```

